

A Multiclass Financial Transaction Fraud Prediction Scheme Using Machine Learning, Ensemble feature Selection, and XAI Technique

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Abstract—Predicting financial transaction fraud is crucial since it costs both consumers and businesses a lot of money. Prior research on predicting financial transaction fraud has only used binary classification. Lower accuracy, improper data preparation, annotation, feature engineering, hyper parameter selection, and model interpretation outcomes are some of the problems with the earlier work. By taking into account ATO, phishing, bank account, refund, and card skimming fraud, this study offers a multiclass financial transaction fraud prediction technique based on machine learning to address the previous issues. By comparing its performance with five machine learning classifiers, the LGBM model with 99.8% accuracy and 96.5% F1 score is chosen as an appropriate prediction model. Advanced data annotation, preprocessing, SHAP-based model interpretation, SMOTE NC based class imbalance reduction, ensemble feature selection, and optunabased hyper parameter tuning approaches were all employed in proposed LGBM prediction model. The accuracy and F1 score of the proposed LGBM model are 2.8% higher than those of previous studies.

Index Terms—Financial Transaction Fraud, ATO, Machine Learning, Ensemble Feature Selection, Optuna, LGBM.

I. INTRODUCTION

Any deliberate act of dishonesty or criminal activity during a financial transaction to obtain funds, products, or services for illicit personal benefit is known as financial transaction fraud [1]. There are many types of financial transaction fraud. The notable fraud related to financial transaction includes counterfeit credit card, card skimming, identity theft, account takeover, phishing, refund, check, and wire fraud, among others. The rise in online purchasing tools, mobile internet, and e-commerce has led to a recent surge in credit card theft [2]. Through phishing, skimming, and information sharing, credit card information can be accessed without authorization in order to make purchases or conduct transactions. Five

transactions out of every 10,000 credit card transactions are fraudulent, according to [3]. In 2020, credit card fraud accounted for nearly 60 percent of all financial fraud. The work in [4] estimates that credit card fraud would cost the world more than USD 35 billion in 2023. Financial transaction fraud prevention requires sophisticated machine learning systems or ML models for fraud detection, reliable payment system, and adaptive security enhancement systems. Various rule-based and statistical techniques were employed in traditional research [5]. However, these traditional systems can not able to swiftly and effectively recognize new and distinct types of fraud in large amounts of data. Because ML and DL models can process enormous amounts of complicated financial fraud data quickly, they are currently commonly employed for financial fraud detection. The article in [6] predicted financial statement fraud using the extreme gradient boosting (XGB) method. The light GBM algorithm was employed in [7] to identify ATO fraud in fintech businesses. The isolation forest technique is used by the authors in [8] to identify fraudulent UPI transactions in India. The study in [9] created a phishing email detection model based on the SVM algorithm. In [10], the authors use logistic regression and isolation forest techniques to create a system for detecting vehicle insurance fraud. In order to discover anomalies in financial transactions and analyze patterns, the work in [11] employed LLM and data analytics tools. To identify the sort of financial fraud in the Telegram ad messages, the authors in [12] employed multimodal learning. The CNN approach is used in the work in [13] to identify bit coin transaction fraud. The earlier works did not build ML-based multiclass financial fraud prediction schemes by investing multiple fraud types (e.g., ATO, bank fraud, phishing, card skimming, refund fraud). For the best

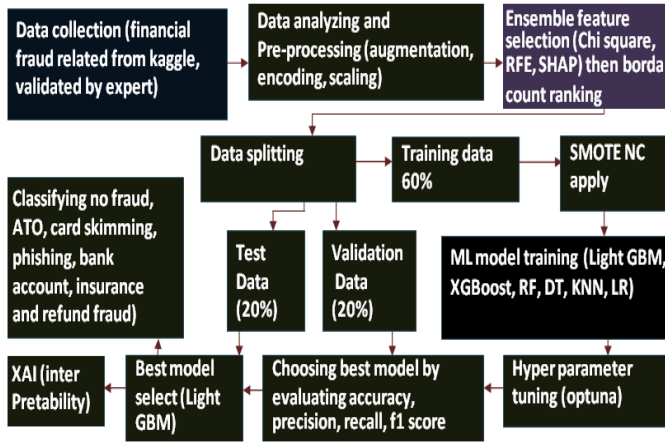


Fig. 1. Proposed methodology for financial payment fraud prediction

feature selection, they didn't use the ensemble feature selection strategy. Proper ML model selection, hyper parameter tweaking, data set annotation, proper data preprocessing techniques are all lacking in the earlier research. The prior research is primarily restricted to binary classification, and its prediction accuracy is subpar. Using ML technique, ensemble feature selection, optuna based hyper parameter tuning, and SHAP-based model interpretation techniques, this paper proposes a multiclass financial transaction fraud prediction scheme. The following are the paper's succinct contributions:

- This study gathers a dataset of financial fraud transactions from Kaggle and uses expert opinions to annotate it for multiclass fraud prediction.
- This study employed sophisticated data preprocessing and ensemble feature selection techniques. By examining RFE, chi square, and SHAP approaches in conjunction with borda count aggregation, the appropriate features for multiclass financial fraud prediction are identified. In this work, class imbalance is resolved using the SMOTE NC approach, while hyper parameter tuning is accomplished using Optuna method.
- Six ML classifiers performances are examined in order to select optimal ML classifier for multiclass financial fraud prediction. The Light GBM is chosen for multi-class financial transaction fraud prediction because it offers the top most prediction accuracy along with top most F1 score value.
- The accuracy and confusion matrix performance of the proposed Light GBM models are contrasted with previous research. This paper presents the SHAP-based feature impact results on models output.

The next section outlined existing research. The multiclass financial fraud prediction system with steps is displayed in Section three. The comparison results are presented in fourth section. A brief synopsis of the paper is included in the last section.

User index	Credit Card use	Year	Amount (usd)	Use chip or online	Merchant name, city	Merchant state, zip	MCC, time	Error occur	Unusual link, mail, device, ip, website, phone trick
1445	Yes	2008	15	chip	Henry, dallas	TX, 78001	4810, 7.1 am	no	no
1460	yes	2009	255	online	Samir, online	NV	5020, 8.5am	yes	yes
1647	yes	2010	60	online	Tom, online	NV	5070, 12.2pm	yes	no
1848	yes	2010	80	swipe	Hary, utah	UT, 84001	6010, 2.12pm	yes	no
1949	yes	2010	125	online	Bun, online	NV	6330, 3.3pm	yes	no
1998	yes	2011	500	online	Rana, online	NV	6550, 4.6pm	yes	no

(a)

Pin capture fallback to swipe	Change pin, password, sus. login	Quick purchase, new device ip for purchase, unusual transaction	Fake signature, fake check, identity theft, trick wire pay	Fake receipt, evidence, claim for refund	Fake insurance claim	Financial Fraud type	Validation
Yes	no	no	no	no	no	No fraud	Yes
no	no	yes	no	no	no	Phishing	Yes
yes	yes	yes	no	no	no	ATO	Yes
yes	no	yes	no	no	no	Card skimming	Yes
no	no	yes	yes	no	no	Bank account fraud	Yes
no	no	no	no	yes	yes	Insurance, refund	Yes

(b)

Fig. 2. (a) Dataset examples (part one), (b) Dataset examples (part two)

Feature name	Description	Features	Description
Credit card use	Whether used credit card or not	Year, time, error	Transaction year, time, error occurred or not
Use chip or online platform	Whether use EMV card or online platform	Merchant name city	Name of business holder, location
Amount transaction	Amount in USD	Merchant state and zip	State and zip code merchant location
MCC	Merchant category business code	Unusual link, mail, website	Whether use fake link, mail, website for tricks
Unusual device, IP use	Abnormal device, ip, phone use for transaction	Pin capture, fallback to swipe	Capture card pin data, use magnetic stripe for transaction
Change pin, password, address, suspicious login	Captured card data then changed pin password by fraudster, multiple failed attempt for login	Quick purchase, unusual transaction	Quick use of card for purchase by fraudster, abnormal type of transaction data like amount, place
Fake signature cheque, evidence, receipt, refund, insurance claim	Used fake data for bank account transaction, fake evidence, receipt refund, fake claim	Fraud type labeling	ATO, card skimming, phishing, bank account fraud, insurance claim fraud, no fraud type

Fig. 3. Dataset features name and description

II. LITERATURE REVIEW

The discussion of the associated literature review is found in section two. The study in [14] achieved more than 90 percent accuracy rate in detecting financial transaction fraud by using SVM and data mining approaches. They were only effective against three kinds of fraud. The study in [15] predicted corporate financial fraud using the PUK-SVM algorithm. The work in [16] uses CNN and Bi-GRU algorithms to forecast master card theft. The article in [17] uses the cat boost algorithm to predict health insurance fraud. The article in [18] deployed quantum computing inspired theory for payment

TABLE I
COMPARISON AND EVALUATION RESULTS

Scheme	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
B. Li et al., [6], XGB	96.90	94	90	92
Biswas et al., [14], SVM	95.20	91.60	90.70	91.10
Chitteti et al., [17], CatBoost	93.00	92.00	95.00	93.70
Proposed work, Light GBM	99.81	99.40	93.90	96.57

Class category	Before SMOTENC	After SMOTENC
ATO (acc. takeover)	6950	10000
Card skimming	8560	10000
Phishing	6800	10000
Bank account fraud	5700	10000
IF or Refund fraud	245	10000
No fraud	892745	892745
Total selected	921000	942745

(a)

Features	Chi square	RFE rank	SHAP rank	Ensemble
Unusual link, email, ip, device use	1	1	1	1
Use chip or online platform	3	2	2	2
Password changed, pin captured, fallback to swipe, identity theft	2	3	3	3
Quick purchase, unusual transaction, suspicious login	4	4	4	4
Fake signature, cheque, fake evidence, tricks for wire transfer	5	5	5	5
Fake receipt, fake claim for refund	6	7	6	6
Amount in transaction	7	6	7	7
Merchant location, name, state, zip code	8	8	8	8
MCC, error occurred	10	9	9	9
Year, time	9	10	10	10

(b)

Fig. 4. (a) Class count after SMOTENC, (b) Feature ranking after ensemble technique

fraud detection. In order to detect UPI fraud in India, the authors of [19] used the Hybrid Markov learning technique. The work in [20] uses computer vision techniques (GAN and CNN) to forecast anomalies (counterfeit checks, or forgeries) in financial transactions. The article in [21] achieved a 73.1 percent accuracy rate in predicting credit card fraud using an unsupervised learning strategy (PSO and K means clustering). In order to forecast financial theft in bit coin networks, the paper in [22] employed ensemble deep learning algorithms. In order to forecast fraudulent activity and transactions in financial institutions, the work in [23] employed a transformer and created knowledge distillation architecture. The XGBoost algorithm was used in the paper in [24] to detect fraud in mobile transactions. For the purpose of managing financial

Light GBM parameters	Values
n_estimators, class	200, 6
Min child	20
Learning rate, leaves	.04, 40
Subsample, depth	.90, 6
Colsample by tree	.90
Reg alpha, lambda	.5, .8

(a)

True level	ATO	1387	0	0	0	0	3
	skimming (CK)	0	1460	0	0	0	252
	phishing (Ph)	0	0	1356	0	0	4
	bank acc. (BA)	0	0	1	1081	0	58
	Ins. refund (IR)	0	0	0	0	41	8
	no fraud (NF)	2	8	0	2	1	178536
	predicted level	ATO	CK	Ph	BA	IR	NF

(b)

Fig. 5. (a) Hyper parameter values (Light GBM), (b) Confusion matrix (Light GBM)

risk in Chinese institutions, the authors in [25] employed both RAG procedures and LLM models. NLP and generative AI techniques were employed in the work in [26] to detect financial fraud. The study in [27] detected accounting fraud by using Knowledge Graphs and Attention Networks. The work in [28] uses DCNN-LSTM-AE-AM models to forecast financial statement fraud in public companies. The existing works mainly considers binary classification (either fraud or non fraud). The proposed work displays a multiclass financial transaction fraud prediction system based on ML, ensemble feature selection, and optuna method based tuning.

III. PROPOSED SCHEME

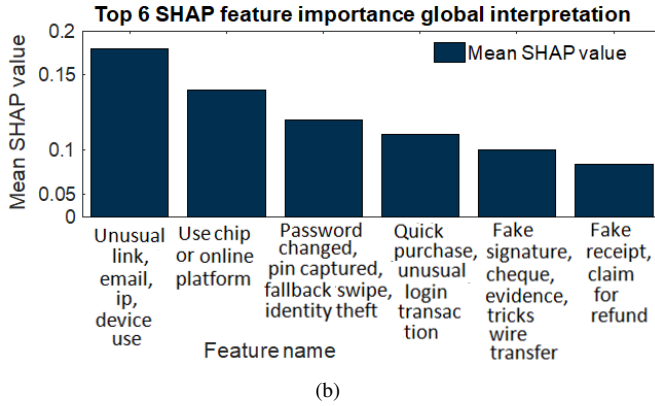
The methodology regarding multiple types of financial transaction fraud prediction using ML and XAI will be depicted in third section.

A. System Model

Figure 1 hints the methodology of proposed multiclass financial transaction prediction model. This paper first gathers

Model	Accuracy	Precision	Recall	F1 score
Light GBM	.998	.994	.939	.965
XGBoost	.985	.91	.89	.90
RF	.980	.89	.86	.87
DT	.975	.88	.84	.86
KNN	.960	.90	.80	.85
LR	.905	.78	.76	.77

(a)



(b)

Fig. 6. (a) Model comparison, (b) SHAP analysis (features)

data on financial transaction fraud from Kaggle [29] and bank officials. Using the recommendations of five experts (i.e., bank officials from City Bank and Sonali Bank), the collected dataset is validated and labeled for multiclass financial fraud detection. Data cleaning, augmentation, transformation, encoding, and scaling operations are then used to preprocess the raw dataset. The optimal feature selection for model training is achieved by ensemble feature selection followed by borda count ranking. Training, testing, and validation sets are created from the entire dataset. In addition optuna-based hyper parameter tuning is used in this paper along with k cross validation. SMOTE NC is utilized to address the class balance problem. To find out the best prediction model, six ML classifiers are tested, including Light GBM, RF, LR, XGBoost, decision tree, and KNN. The best ML model for multiclass financial fraud prediction is selected based on their top most prediction accuracy value. SHAP is used to explain the significance of features in model output.

B. Dataset Preparation

The financial payment fraud dataset is initially gathered via Kaggle for this project. There are over 20 million data points in the original dataset. However, in order to create a suitable dataset with several categories of financial fraud (such as phishing, account takeover or ATO, bank account fraud, card skimming, refund fraud, and no fraud), we chose 921 thousand data entries at random for this study. Information about card use, transaction amount, date, time, merchant name, city, fraud type, susceptible behavior, device id, IP use, and

error information are all included in each dataset record. An example of dataset entries is shown in Figure 2. During the dataset validation process, bank sector experts integrate certain feature sets along with the goal class level. Figure 3 shows the dataset features name and explanation. Experts use majority vote to resolve ambiguities throughout the dataset annotation process. Figure 4(a) displays the financial payment fraud class record before to and following SMOTE NC.

C. Preprocessing

Thorough data preprocessing procedures, including handling missing values and removing empty values, were applied to the original dataset. After that, categorical data (such as merchant city and merchant state) is encoded using a label encoder. Label encoders are used to convert categorical data into numerical representations. Numerical values, such as 1 or 0, are used to represent binary attributes, such as altered password or pin. Numerical features (e.g., device id, website link) were subjected to a robust scalar regarding outliers handling. Using IQR range, robust scalar scales the data and eliminates the median.

D. Feature Selection

For the optimal feature selection and ranking, this work employed the chi square, RFE, and SHAP techniques. The scores produced by several feature selection techniques were combined in this paper using the Borda count aggregation technique. Ultimately, the ensemble feature selection technique and borda count rank are used to choose the best features. Figure 4(b) shows the list of the top ten features that were chosen. The unusual link, device, IP, and link feature used for the transaction secured the top spot, as shown in Figure 4(b). Using a chip or online platform for transactions is the second crucial feature. Identity theft, pin capture, password change, and fallback swipe are the third crucial feature. Rapid purchases, odd transactions, and suspicious login attempts rank as the fourth crucial feature for fraud prediction.

E. Model Selection, Tuning

The optuna-based hyper parameter tuning parameters for the Light GBM approach is first displayed in the Figure 5(a). The LGBM models true positive and false positive value is incorporated in the next (Figure 5(b)). The comparison of the ML models is tabulated in the next (Figure 6(a)). With 99.8 percent accuracy and 96.5 percent F1 score, the Light GBM model outperforms all other ML schemes in multiclass financial fraud prediction, according to Figure 6(a). This is because; compared to all other models, the Light GBM model has a greater true positive rate and a lower false negative rate. For large datasets and high dimensional features, LGBM performs better than XGBoost. The six most significant feature based on SHAP analysis is shown in Figure 6(b) along with their effect on model output. It can be demonstrated that the feature that impacts highest on model output (mean SHAP value of .16) is unusual link, IP, and device usage.

IV. RESULTS AND ANALYSIS

The evaluation results of the previous studies for multiclass financial fraud prediction and our proposed LGBM technique is hinted in Table I. The proposed Light GBM model outperforms the other examined approaches (i.e., [6], [14], and [17]) in terms all performance metrics. Compared to works in [6], [17], and [14], the proposed Light GBM model provides at least 2.8 percent more F1 score and accuracy. Because of its adaptive data preprocessing, SMOTE NC-based class imbalance solving, ensemble feature selection, optuna-based tuning, cross validation XAI technique, and multiclass labeling and validation, the LGBM model is better for multiclass financial fraud prediction.

V. CONCLUSION

A financial transaction fraud prediction technique based on ML and explainable AI is presented in the study. The proposed prediction scheme can identify ATO, phishing, card skimming, bank account, refund, and no fraud classes. This work gathers financial transaction fraud data from Kaggle and uses bank experts' comments to annotate the data for multiclass financial transaction fraud detection. Preprocessing and SMOTE NC technology is employed to overcome class imbalance issue. Ensemble feature selection is used to find the optimal features for model training. Optuna is employed for hyper parameter tuning. Among the tested ML classifiers, LGBM model has the greatest accuracy of 99.8% and F1 score of 96.5%, it is chosen for multiclass financial fraud prediction. The LGBM model provides at least 2.8 percent more accuracy and F1 score than previous models. Future research could include, among other things, FL, AI, and transformer-based financial statement fraud detection, block chain based gateway and account verification system, computer vision and AI based fake signature prediction, and LLM and RAG-based financial fraud trend analysis, among others.

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